

Deliverable 1: Intelligence Scope Document

INTELLIGENCE SCOPE DOCUMENT

Document ID: ISD-2026-055-001

Date: September 22, 2026

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Incident: MFG Accounting Workstation Anomaly (ACC-227)

SECTION 1: REQUEST COMPLETENESS AUDIT

Requirement Element	Present? (Y/N)	If Missing, What Is Needed?
Specific threat actor/incident identifier	Y	ACC-227 PC Problem / CrimsonQuill Threat
Timeframe of intelligence needed	Y	2026-05-12 to 2026-05-15 UTC (Past 72 hours)
Geographic jurisdiction(s) involved	N	Jurisdiction not stated matters since C2 infra is Belarus-hosted and customer base may cross borders
Specific data types requested	Y	CISO's questions (what happened, who, what data, what now) frame the request
Allowed collection methods	N	No written authorization for dark web forum data or paid OSINT feeds
Deliverable format	Y	answers needed within 8 hours for the board meeting
Budget constraint	Y	Budget is not a constraint for remediation

SECTION 2: LEGAL RISK IDENTIFICATION

Identified Legal Risk	Relevant Law/Policy	Risk Level (High/Med/Low)	Justification
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Unauthorized active engagement with foreign-hosted C2 infrastructure	Computer Fraud and Abuse Act (CFAA)	High	Any scanning, disruption, or "hack-back" against the Belarus AS6699 infrastructure would exceed our legal authority to act
Unredacted sharing of exfiltrated data during external reporting	GLBA / GDPR / State Privacy Laws	High	Accessed data includes customer investment portfolios and payroll records sharing this raw with ISAC partners or regulators would itself create exposure
Evidence handling affecting later legal/insurance use	Federal Rules of Evidence / internal forensic policy	Medium	If this incident heads toward law enforcement referral or a cyber-insurance claim, chain of custody on the firewall/EDR artifacts needs to hold up

SECTION 3: PRIVACY CONSTRAINT IDENTIFICATION

Privacy Concern	Specific Data Element at Risk	Mitigation Required
Exposure of Customer PII / Financial Records	Customer Investment Portfolios (100k+ users)	Anonymize/redact sensitive PII in reporting; encrypt incident response data handling channels.
Employee PII Disclosure	Employee Payroll Records, jsmith account credentials	Limit internal disclosure on a strict need-to-know basis (TLP:AMBER+STRICT within IR team).

Privacy Concern	Specific Data Element at Risk	Mitigation Required
Exposure of Customer PII / Financial Records	Investment portfolio records, payroll records	Anonymize or hash customer/employee identifiers before any report leaves the internal team
Exposure of confidential deal and governance material	M&A target file, board meeting minutes	Reference these only by filename and hash in external-facing IoC feeds never their contents

SECTION 4: EXPRESS SCOPE DEFINITION

In Scope (what WILL be collected/methods WILL be used):

1. Passive review of firewall, EDR, and internal log artifacts from ACC-227
2. Correlation of extracted indicators against OSINT feeds (abuse.ch) and WHOIS data
3. Passive reading of already-collected dark web forum content
4. Structured ACH hypothesis testing on attacker attribution
5. Production of MISP and STIX deliverables plus containment recommendations for the technical team

Out of Scope (explicitly forbidden or excluded):

1. Active scanning, disruption, or hack-back against AS6699 infrastructure
2. Direct contact with threat actors (e.g., messaging CrimsonTrader)
3. Public release of unredacted customer PII or internal corporate documents

SECTION 5: SIGNED APPROVAL

Analyst Certification: I have performed an ethical and legal audit of the CISO's request dated 2026-05-15. The scope defined above complies with applicable laws and organizational policies.

Signed: Raul Austero Varias JR.

Date: September 22, 2026

Recommended Action (Check one):

☒ Proceed with scope as defined above

☐ Return to requestor for clarification (specify what is missing)

☐ Reject entirely (specify legal/ethical barrier that cannot be resolved)

SECTION 2: COLLECTION & ANALYSIS

Deliverable 2: Extracted and Normalized IoCs

#	Type	Value	Source (Evidence Letter)	First Seen	Confidence (0-100)
1	IP	45.155.88.33	Evidence A, B, C	2026- 05-12	100%
2	IP	45.155.88.34	Evidence A, B, C	2026- 05-13	95%
3	Domain	finance-update-portal.com	Evidence C, D	2026- 05-10	85%
4	URL	http://45.155.88.33/loader.ps1	Evidence B, C	2026- 05-12	95%
5	URL	http://45.155.88.33/mimikatz.exe	Evidence B, C	2026- 05-12	95%
6	Filename	update.exe	Evidence B	2026- 05-12	90%
7	Filename	mimikatz.exe	Evidence B	2026- 05-13	90%
8	MD5	d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9	Evidence B	2026- 05-12	85%
9	URL	http://45.155.88.34/update.bin	Evidence C	2026- 05-13	90%

Part B: Normalization

Convert the raw firewall logs (Evidence A) into normalized JSON format. Extract ONLY records involving external IPs (exclude internal destination 10.22.45.12). Use the schema below.

Normalized JSON Output:

```
{
  "normalized_network_events": [
    {
      "timestamp": "2026-05-12T13:22:15Z",
      "source_ip": "10.22.35.104",
      "destination_ip": "45.155.88.33",
      "bytes_out": 1524,
      "protocol": "TCP"
    },
    {
      "timestamp": "2026-05-12T13:22:18Z",
      "source_ip": "10.22.35.104",
      "destination_ip": "45.155.88.33",
      "bytes_out": 8230,
      "protocol": "TCP"
    },
    {
      "timestamp": "2026-05-13T02:15:33Z",
      "source_ip": "10.22.35.104",
      "destination_ip": "45.155.88.33",
      "bytes_out": 125000,
      "protocol": "TCP"
    },
    {
      "timestamp": "2026-05-13T02:22:44Z", "source_ip": "10.22.35.104",
      "destination_ip": "45.155.88.33", "bytes_out": 624,
      "protocol": "TCP"
    },
    {
      "timestamp": "2026-05-14T02:18:22Z", "source_ip": "10.22.35.104",
      "destination_ip": "45.155.88.34", "bytes_out": 856,
      "protocol": "TCP"
    },
    {

```

```

"timestamp": "2026-05-14T03:00:33Z", "source_ip": "10.22.35.104",
"destination_ip": "45.155.88.33", "bytes_out": 2450000,
"protocol": "TCP"
},
{
"timestamp": "2026-05-15T02:10:15Z", "source_ip": "10.22.35.104",
"destination_ip": "45.155.88.34", "bytes_out": 890,
"protocol": "TCP"
},
{
"timestamp": "2026-05-15T02:45:22Z", "source_ip": "10.22.35.104",
"destination_ip": "45.155.88.33", "bytes_out": 3120000,
"protocol": "TCP"
}
]
}

```

Part C: Internal Data Capture

From the EDR logs (Evidence B), extract the following:

1. The user account that was compromised: MFG\jsmith
2. The initial infection vector (parent process of powershell.exe): outlook.exe
3. The malware family name (from OSINT): CrimsonQuill
4. The internal file server IP that was accessed: 10.22.45.12
5. The total data exfiltrated (sum of bytes_out from firewall logs for external IPs): 0.0057 GB (5,706,124 bytes recorded across active technical log fragments, within a broader multi-day payload total up to 8.7 GB).

SECTION 3: PATTERN RECOGNITION & HYPOTHESIS

Deliverable 3: Pattern Analysis and Competing Hypotheses

Part A: Time of Day Pattern Analysis

From Evidence A (firewall logs), analyze the timing of external connections (to 45.155.88.33 and 45.155.88.34):

Hour Range (UTC)	Number of Connections	Percentage
00:00-05:59	6	75%

06:00-11:59	0	0%
12:00-17:59	2	25%
18:00-23:59	0	0%

What pattern do you observe? (One sentence): Nearly all C2 activity clusters in the 02:00–04:00 UTC window, well outside normal Accounting department working hours, consistent with automated beaconing rather than manual daytime use.

Part B: Tool Choice Pattern Analysis

From Evidence B (EDR logs), list the sequence of tools used in order (starting from initial access):

Tool Chain:

1. outlook.exe (Initial access attachment vector)
2. powershell.exe (Network download cradle execution)
3. update.exe (Localized backdoor setup and foothold maintenance)
4. svchost.exe (Masqueraded network check-in and persistence loops)
5. mimikatz.exe (LSASS parsing for administrative credential harvesting)

What is the most common 2-tool combination? powershell.exe → mimikatz.exe

Part C: Competing Hypotheses

Write THREE competing hypotheses for the attack's origin. Use the ACH scoring matrix.

Hypothesis H1 (Single Actor): One actor purchased CrimsonQuill from CrimsonTrader, ran the phishing campaign personally, and carried out the entire intrusion end-to-end without outside help.

Hypothesis H2 (MaaS/Multiple Actors): A buyer obtained access credentials separately from DataBroker_44 and deployed the purchased CrimsonQuill toolkit against that access — two different actors playing two different roles.

Hypothesis H3 (Insider Threat): An MFG accounting employee possibly jsmith intentionally ran the tooling and moved data to the attacker infrastructure themselves.

ACH Scoring Matrix: Score each hypothesis against the evidence below (-2 to +2)

Evidence	H1 (Single Actor)	H2 (MaaS)	H3 (Insider)
Dark web: CrimsonTrader sells tool	+1	+2	-1
Dark web: DataBroker_44 sells access	-1	+2	-1
Dark web: UnknownUser claims credit	0	+1	0
Timing: Consistent 02:00-04:00 UTC	+1	+1	-2
Tools: PowerShell + Mimikatz pattern	+1	+1	-1
Data accessed: M&A + SWIFT + customer	0	+1	+1
TOTAL SCORE	2	8	-4

Ranking (1st to 3rd):

1st: H2 – Score: +8

2nd: H1 – Score: +2

3rd: H3 – Score: -4

Final Analytical Judgment (one sentence):

The dark web forum activity a separate tool seller and a separate access broker, each speaking to a different piece of the operation combined with the automated off-hours beaconing pattern, points most strongly to a MaaS-style intrusion carried out by a buyer using purchased access and a purchased toolkit, rather than one actor working alone or an insider acting from within.

SECTION 4: MISP ENRICHMENT

Deliverable 4: MISP Event Creation

```
{
  "Event": {
    "info": "CrimsonQuill Intrusion and Data Exfiltration - Meridian Financial Group (ACC-227)",
    "date": "2026-05-12",
    "threat_level_id": 1,
    "analysis": 2,
    "distribution": 1,
    "tags": [
      {"name": "tlp:amber", "colour": "#FFC000"},
    ]
  }
}
```

```

{"name": "malware:family=\"CrimsonQuill\"", "colour": "#FF0000"},
{"name": "target:sector=\"Financial\"", "colour": "#00308F"},
{"name": "verification:confirmed", "colour": "#008000"}
],
"Attribute": [
  {"type": "ip-dst", "category": "Network
activity", "value": "45.155.88.33", "to_ids": true, "comment": "Primary C2, hosted on AS6699
(Belarus)", "first_seen": "2026-05-
12", "tags": [{"name": "confidence:100", "colour": "#0000FF"}, {"name": "source:type=\"Firewall\"",
"colour": "#FFA500"}]},
  {"type": "ip-dst", "category": "Network
activity", "value": "45.155.88.34", "to_ids": true, "comment": "Secondary/backup
C2", "first_seen": "2026-05-
13", "tags": [{"name": "confidence:95", "colour": "#0000FF"}, {"name": "source:type=\"OSINT\"",
"colour": "#FFA500"}]},
  {"type": "domain", "category": "Network activity", "value": "finance-update-
portal.com", "to_ids": true, "comment": "Phishing domain, bulletproof
nameservers", "first_seen": "2026-05-
10", "tags": [{"name": "confidence:85", "colour": "#0000FF"}, {"name": "source:type=\"OSINT\"",
"colour": "#FFA500"}]},
  {"type": "url", "category": "Network
activity", "value": "http://45.155.88.33/loader.ps1", "to_ids": true, "comment": "PowerShell
download cradle target", "first_seen": "2026-05-
12", "tags": [{"name": "confidence:95", "colour": "#0000FF"}, {"name": "source:type=\"EDR\"", "col
our": "#FFA500"}]},
  {"type": "url", "category": "Network
activity", "value": "http://45.155.88.33/mimikatz.exe", "to_ids": true, "comment": "Credential-theft
tool download", "first_seen": "2026-05-
12", "tags": [{"name": "confidence:95", "colour": "#0000FF"}, {"name": "source:type=\"EDR\"", "col
our": "#FFA500"}]},
  {"type": "url", "category": "Network
activity", "value": "http://45.155.88.34/update.bin", "to_ids": true, "comment": "Secondary payload
host", "first_seen": "2026-05-
13", "tags": [{"name": "confidence:90", "colour": "#0000FF"}, {"name": "source:type=\"OSINT\"",
"colour": "#FFA500"}]},

```

```

    {"type": "filename", "category": "Payload
delivery", "value": "update.exe", "to_ids": true, "comment": "Dropped loader establishing
foothold", "first_seen": "2026-05-
12", "tags": [{"name": "confidence:90", "colour": "#0000FF"}, {"name": "source:type=\\EDR\\", "col
our": "#FFA500"}]},

    {"type": "filename", "category": "Payload
delivery", "value": "mimikatz.exe", "to_ids": true, "comment": "Credential dumping
utility", "first_seen": "2026-05-
13", "tags": [{"name": "confidence:90", "colour": "#0000FF"}, {"name": "source:type=\\EDR\\", "col
our": "#FFA500"}]},

    {"type": "md5", "category": "Payload
delivery", "value": "d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9", "to_ids": true, "comment": "Hash of the
update.exe loader", "first_seen": "2026-05-
12", "tags": [{"name": "confidence:85", "colour": "#0000FF"}, {"name": "source:type=\\EDR\\", "col
our": "#FFA500"}]}

]

}

}

```

SECTION 5: STIX REPORTING

Deliverable 5: STIX 2.1 Bundle

```

{
  "type": "bundle",
  "id": "bundle--3d4e5f6a-7b8c-4d9e-8f0a-1b2c3d4e5f6a",
  "spec_version": "2.1",
  "objects": [
    {
      "type": "identity",
      "spec_version": "2.1",
      "id": "identity--a1b2c3d4-e5f6-4a7b-8c9d-0e1f2a3b4c5d",
      "created": "2026-05-15T12:00:00Z",

```

```
"modified": "2026-05-15T12:00:00Z",
"name": "Meridian Financial Group",
"identity_class": "organization",
"sectors": ["financial-services"]
},
{
  "type": "malware",
  "spec_version": "2.1",
  "id": "malware--b2c3d4e5-f6a7-4b8c-9d0e-1f2a3b4c5d6e",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "name": "CrimsonQuill",
  "malware_types": ["trojan"],
  "is_family": true,
  "description": "Banking trojan sold as a kit, featuring a PowerShell download cradle with AMSI bypass, Mimikatz-based credential theft, HTTPS C2, and automated exfiltration of financial documents."
},
{
  "type": "threat-actor",
  "spec_version": "2.1",
  "id": "threat-actor--c3d4e5f6-a7b8-4c9d-0e1f-2a3b4c5d6e7f",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "name": "Unidentified CrimsonQuill Operator",
  "threat_actor_types": ["criminal"],
  "primary_motivation": "financial-gain",
```

"description": "Buyer of access from DataBroker_44 and tooling from CrimsonTrader, assessed as the operator behind the MFG intrusion."

},

{

"type": "campaign",

"spec_version": "2.1",

"id": "campaign--d4e5f6a7-b8c9-4d0e-1f2a-3b4c5d6e7f8a",

"created": "2026-05-15T12:00:00Z",

"modified": "2026-05-15T12:00:00Z",

"name": "MFG CrimsonQuill Breach",

"objective": "Theft of SWIFT wire logs, M&A documents, and customer portfolio data from Meridian Financial Group",

"first_seen": "2026-05-12T13:21:05Z",

"last_seen": "2026-05-15T02:45:22Z"

},

{

"type": "attack-pattern",

"spec_version": "2.1",

"id": "attack-pattern--e5f6a7b8-c9d0-4e1f-2a3b-4c5d6e7f8a9b",

"created": "2026-05-15T12:00:00Z",

"modified": "2026-05-15T12:00:00Z",

"name": "Phishing: Spearphishing Attachment",

"external_references": [{"source_name": "mitre-attack", "external_id": "T1566.001"}]

},

{

"type": "attack-pattern",

"spec_version": "2.1",

"id": "attack-pattern--f6a7b8c9-d0e1-4f2a-3b4c-5d6e7f8a9b0c",

```
"created": "2026-05-15T12:00:00Z",
"modified": "2026-05-15T12:00:00Z",
"name": "Command and Scripting Interpreter: PowerShell",
"external_references": [{"source_name": "mitre-attack", "external_id": "T1059.001"}]
},
{
  "type": "attack-pattern",
  "spec_version": "2.1",
  "id": "attack-pattern--a7b8c9d0-e1f2-4a3b-4c5d-6e7f8a9b0c1d",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "name": "OS Credential Dumping: LSASS Memory",
  "external_references": [{"source_name": "mitre-attack", "external_id": "T1003.001"}]
},
{
  "type": "indicator",
  "spec_version": "2.1",
  "id": "indicator--b8c9d0e1-f2a3-4b4c-5d6e-7f8a9b0c1d2e",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "pattern": "[ipv4-addr:value = '45.155.88.33']",
  "pattern_type": "stix",
  "valid_from": "2026-05-12T13:22:15Z",
  "confidence": 100
},
{
  "type": "indicator",
```

```
"spec_version": "2.1",
"id": "indicator--c9d0e1f2-a3b4-4c5d-6e7f-8a9b0c1d2e3f",
"created": "2026-05-15T12:00:00Z",
"modified": "2026-05-15T12:00:00Z",
"pattern": "[ipv4-addr:value = '45.155.88.34']",
"pattern_type": "stix",
"valid_from": "2026-05-13T08:15:00Z",
"confidence": 95
},
{
  "type": "indicator",
  "spec_version": "2.1",
  "id": "indicator--d0e1f2a3-b4c5-4d6e-7f8a-9b0c1d2e3f4a",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "pattern": "[domain-name:value = 'finance-update-portal.com']",
  "pattern_type": "stix",
  "valid_from": "2026-05-10T10:22:15Z",
  "confidence": 85
},
{
  "type": "indicator",
  "spec_version": "2.1",
  "id": "indicator--e1f2a3b4-c5d6-4e7f-8a9b-0c1d2e3f4a5b",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "pattern": "[url:value = 'http://45.155.88.33/loader.ps1']",
```

```
"pattern_type": "stix",
"valid_from": "2026-05-12T13:21:05Z",
"confidence": 95
},
{
  "type": "indicator",
  "spec_version": "2.1",
  "id": "indicator--f2a3b4c5-d6e7-4f8a-9b0c-1d2e3f4a5b6c",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "pattern": "[file:hashes.'MD5' = 'd4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9']",
  "pattern_type": "stix",
  "valid_from": "2026-05-12T13:21:10Z",
  "confidence": 85
},
{
  "type": "course-of-action",
  "spec_version": "2.1",
  "id": "course-of-action--a3b4c5d6-e7f8-4a9b-0c1d-2e3f4a5b6c7d",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "name": "Isolate host and rotate exposed credentials",
  "description": "Network-isolate ACC-227 via EDR; block the identified C2 IPs and domain
at the firewall/DNS layer; force a password reset for jsmith and the wider Accounting
department; reissue any credentials that touched SWIFT or wire systems."
},
{
  "type": "course-of-action",
```

```
"spec_version": "2.1",  
"id": "course-of-action--b4c5d6e7-f8a9-4b0c-1d2e-3f4a5b6c7d8e",  
"created": "2026-05-15T12:00:00Z",  
"modified": "2026-05-15T12:00:00Z",  
"name": "Add detection for PowerShell download cradles and LSASS access",  
"description": "Deploy EDR rules for IEX/WebClient download-cradle behavior, AMSI  
bypass attempts, and non-standard processes reading LSASS memory (the Mimikatz sekurlsa  
pattern)."
```

```
},
```

```
{
```

```
"type": "relationship",  
"spec_version": "2.1",  
"id": "relationship--c5d6e7f8-a9b0-4c1d-2e3f-4a5b6c7d8e9f",  
"created": "2026-05-15T12:00:00Z",  
"modified": "2026-05-15T12:00:00Z",  
"relationship_type": "uses",  
"source_ref": "threat-actor--c3d4e5f6-a7b8-4c9d-0e1f-2a3b4c5d6e7f",  
"target_ref": "malware--b2c3d4e5-f6a7-4b8c-9d0e-1f2a3b4c5d6e"
```

```
},
```

```
{
```

```
"type": "relationship",  
"spec_version": "2.1",  
"id": "relationship--d6e7f8a9-b0c1-4d2e-3f4a-5b6c7d8e9f0a",  
"created": "2026-05-15T12:00:00Z",  
"modified": "2026-05-15T12:00:00Z",  
"relationship_type": "attributed-to",  
"source_ref": "campaign--d4e5f6a7-b8c9-4d0e-1f2a-3b4c5d6e7f8a",  
"target_ref": "threat-actor--c3d4e5f6-a7b8-4c9d-0e1f-2a3b4c5d6e7f"
```

```
},
{
  "type": "relationship",
  "spec_version": "2.1",
  "id": "relationship--e7f8a9b0-c1d2-4e3f-4a5b-6c7d8e9f0a1b",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "relationship_type": "targets",
  "source_ref": "campaign--d4e5f6a7-b8c9-4d0e-1f2a-3b4c5d6e7f8a",
  "target_ref": "identity--a1b2c3d4-e5f6-4a7b-8c9d-0e1f2a3b4c5d"
},
{
  "type": "relationship",
  "spec_version": "2.1",
  "id": "relationship--f8a9b0c1-d2e3-4f4a-5b6c-7d8e9f0a1b2c",
  "created": "2026-05-15T12:00:00Z",
  "modified": "2026-05-15T12:00:00Z",
  "relationship_type": "indicates",
  "source_ref": "indicator--b8c9d0e1-f2a3-4b4c-5d6e-7f8a9b0c1d2e",
  "target_ref": "malware--b2c3d4e5-f6a7-4b8c-9d0e-1f2a3b4c5d6e"
}
]
}
```